The Design of a Hip Exoskeleton and a Review of Its Adverse Effects on Gait

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Abstract

Objective:

This literature review will analyze design considerations regarding hip-attached exoskeletons and their potential adverse effects. Individuals with degenerative diseases and other forms of disability often experience significant gait impairment; while modern robotic technologies can assist with these challenges, limited research has been conducted to ensure the safety and efficacy of long-term use[1], [2]. Thus, this literature review analyzes publications examining design considerations that amplify or mitigate adverse effects. While hip exoskeletons remain in early stages of widespread deployment, they hold considerable potential. Key parameters related to the benefits and limitations of hip exoskeleton use are examined. By synthesizing findings from published studies, this review aims to provide insights into optimizing exoskeleton design and functionality to improve overall clinical effectiveness.

Methods:

This literature review synthesizes eight peer-reviewed sources. Of those eight sources, two are systematic reviews and meta-analyses, five are case studies and clinical trials, and one is a design proposal. The research topic of “Exoskeletons and gait rehabilitation” was initially explored, followed by the question “What are the potential adverse effects of these exoskeletons?” From this path, sources [3]-[6] were utilized to determine the design process of hip exoskeletons. Sources [1] and [2] were used to consider the specific adverse effects that could result from HE use. Sources [7] and [8] were used to identify the gait parameters most directly improved by HE use. Each source was referenced multiple times throughout the review and is not restricted to a single subsection. Sources are organized according to their primary application within the review.

Sources:

The eight sources included in this literature review were drawn from the following peer-reviewed journals. Source[1] appeared in Frontiers in Robotics and AI. Source[2] appeared in the Journal of NeuroEngineering and Rehabilitation. Source[3] appeared in Scientific Reports. Source[4] appeared in Geriatric Orthopaedic Surgery & Rehabilitation. Source[5] appeared in IEEE Transactions on Neural Systems and Rehabilitation Engineering. Source[6] appeared in Actuators. Source[7] appeared in BMC Musculoskeletal Disorders. Source[8] appeared in Gait & Posture. All listed sources are peer-reviewed scholarly journals, each reviewed for content pertaining to gait rehabilitation using robotic devices.

Conclusions:

A substantial body of research supports the benefits of hip exoskeletons for gait rehabilitation in individuals with impaired mobility. However, minimal data exist regarding the potential adverse effects of their use. Although numerous parameters can assess exoskeleton effectiveness, limited work has established statistically significant findings concerning negative outcomes. Future studies must dedicate greater attention to systematically tracking the parameters that adversely affect wearer gait and overall health.

Introduction

Individuals who have experienced neurological disability frequently present with lower-body motor impairment. Of individuals who have suffered from a stroke, more than 80% report some degree of walking dysfunction, with profound consequences for daily locomotion [5]. While these impairments can be addressed through traditional physical therapy, the rehabilitation process is often lengthy and demanding. As technology advances, gait-assisting hip exoskeletons (HEs) have emerged as a promising adjunct to support rehabilitation efforts and restore mobility.

Recent advances have established that "robotic devices can assure consistency in repetitive rehabilitation therapy and are available for rehabilitation treatment" [7]. For individuals with impaired gait, restoring functional mobility is the overarching goal of rehabilitation. Finding the most effective and efficient methods is the leading objective for research in this area. While hip exoskeletons offer numerous demonstrated benefits, limited research has been conducted to determine whether adverse effects arise from prolonged use [1]. Numerous studies have examined the parameters for which hip exoskeletons yield positive outcomes, yet very few have investigated statistically substantial data regarding potential adverse effects. Furthermore, the significance of design variation in relation to prospective adverse events remains poorly understood [1]. This literature review will analyze existing research on hip exoskeletons, focusing on design and outcomes to determine whether any adverse effects are statistically significant.

Body

Hip Exoskeleton Overview

A hip exoskeleton, at least one designed for gait rehabilitation, attaches directly to the wearer's hips and applies supplemental torques when necessary [6]. The Human Robot Systems Laboratory (HRSL) at UMass Amherst is currently studying its own HE to research how the wearer's body reacts to externally applied forces [5]. A typical HE applies targeted forces and torques to assist the wearer in maintaining a functional walking pattern. For individuals with lower-body impairment, the primary goal of HE-assisted therapy is to restore mobility toward a pre-injury or normative baseline.

The HRSL's main goal was to explore the possibilities of future applications for HEs and to determine their significance in gait rehabilitation [5]. In their study, the lab reviewed the effects of an HE's ability to apply bilateral asymmetric stiffnesses to individuals walking on a treadmill. The lab concluded that "Over time, participants adapted towards a more symmetric gait pattern in most kinematic and kinetic gait measures" [5]. This is a similar trial to that of source [4], which reviews the effectiveness of an HE on an individual who recently underwent a Total Knee Arthroplasty (TKA)(knee replacement). The researchers trialed the Honda Walking Assist (HWA) on a 76-year-old woman who recently had TKA. They "assessed walking ability, knee function, and the gait kinematics parameters…" and concluded that "…gait training with the HWA may have contributed to the early improvement in gait after TKA" [4]. The findings of both studies lean toward the conclusions of other publications that HEs have strictly positive effects when used safely and correctly.

One thing to note is that the HRSL conducted their study on thirteen unimpaired individuals, while source [4]'s subject was suffering from gait impairment. This is important to remember, as both groups, impaired and unimpaired participants, yielded comparable results.

Design

Most established hip exoskeletons have differing design specifications between various labs. Typically, HEs contain a rigid frame that houses the actuators, motors, and other dynamic systems that allow the contraption to move. The exoskeleton's control system, usually incorporating various forms of sensors and algorithms, monitors the wearer's movements and can adjust the applied loads in real-time [6]. Some designs, like the HRSL's, prioritize lightweight materials to enhance wearability and comfort [5]. The HRSL's HE has a nearly fully 3D-printed frame, making it extremely lightweight. Others, like the proposed design in source [6], focus on the maximization of potential torque output and precision for power augmentation, with minimal care for overall size [6]. These differences reflect the specific application and intended use case of each HE, whether clinical, industrial, or gait rehabilitation. The Active Pelvis Orthosis (APO) from source [3] utilized a training profile that was implemented specifically to "enhance eccentric activation of hip extensors while providing hip flexor assistance during swing, to reduce the total energetic cost of walking" [3]. Each exoskeleton was built with a different main consideration in mind, allowing for three distinguishable designs.

Several challenges arise when considering the design of an HE. For one, the number of degrees of freedom (DOF) on which the HE can move along. Many labs incorporate a 2 DOF system for each hip joint. This usually allows for "active flexion/extension and passive abduction/adduction" [6]. Some labs have utilized a six-DOF system for their HE, which offers two degrees of freedom for internal and external rotation of each hip joint. This remains a relatively uncommon approach, as noted in source [6]. Most labs have concluded that a four-DOF system offers the most benefits for gait rehabilitation and improvement as "[HEs] should provide flexibility in lower limb dynamics through sufficient degrees of freedom in all three motion planes" [7]. When considering the weight of the HE, the number of the DOF plays a direct impact. The more DOFs that are made active, the heavier the HE will be, as there will be more equipment required to utilize the activeness of the given DOF.

In general, a hip exoskeleton's main purpose is to compensate for impaired muscle function. Actuators and motors apply linear forces and rotational torques to the wearer's joints, resulting in easier motion and reduced physical effort. The sensors incorporated within the design allow for adaptive assistance, personalizing the movement to the wearer's specific needs [1], [3], [5], [7].

Impact On Gait Rehabilitation

As previously mentioned, hip exoskeletons remain in an experimental phase, with ongoing research evaluating whether they can reliably supplement or eventually replace traditional gait rehabilitation methods. Traditional therapies focus on improving gait through specifically targeted exercise and manual assistance from a trainer. These methods are "labor-intensive" and highly susceptible to outcome variability [1], [4], [6], [7]. The implementation of hip exoskeletons has the potential to reduce variability in the outcomes and to enhance consistent rehabilitation.

There are various reasons why an individual would need an HE to apply additional loads to their hips that assist in the walking motion. As already stated, individuals who have suffered from stroke or other neurological disorders often struggle with gait impairment, but they are not the only ones who face this issue. Elderly individuals who have varying cases of arthritis or other degenerative diseases would benefit from supplemental torques being added to their already stiff joints. According to the meta-analysis from source [7], “Gait and balance problems contribute to poor quality of life and increased morbidity and mortality in the elderly” [7]. The elderly subjects examined in most studies, including that of source [3], have undergone traditional therapies, with minimal positive results. This is expected, as most therapies strictly benefit younger, healthier individuals. As the subjects grow in age, HEs are more likely to be the sole contributor to improvements in gait, and according to source [7], “…ambulatory devices are the only option left in most cases…” [7].

The primary reason behind continuing the development of HEs is to improve the parameters directly related to an individual's walking ability. Several parameters are viewed as the most crucial by sources [2], [3], [4], [5], [7], [8]. As worded in source [7] “Analyses [were] performed in terms of gait spatiotemporal parameters like speed (self-speed and maximum speed), step length, stride length, cadence, and oxygen consumption” [7]. Almost all studies that have done some kind of trial of their own used these parameters to determine the effectiveness of their specific HE on the subjects at hand. Comparing the before and after of the given parameters allowed each group of researchers to dictate the benefits of the HE in their experiment. Source [2] saw a .13 m/s increase in walking speed (arguably the most important parameter). Source [7] saw a similar increase, with a range of .10 m/s to .17 m/s in the studies they reviewed in their meta-analysis. Source [8], while focusing on a back-support exoskeleton and not a hip exoskeleton, determined similar results to that of the other studies, verifying a significant increase in both gait speed and preferred walking speed. For subjects’ stride length, source [7] notes an improvement of .06m per step, while the researchers from the HRSL at UMass in source [5] recorded that the use of their HE resulted in “…more symmetric step length enabled by compensations at the knee…” [5]. The HRSL also noted an improved(decreased) rate of energy consumption, which they deemed to be a positive byproduct of their overall goal, to return symmetric gait patterns from asymmetric discrepancies [5]. As for the metabolic cost of walking, HEs can assist with the motions that are the most taxing of the gait cycle – hip flexion and extension. In their research, source [7] found that the “…net cardiopulmonary metabolic energy cost was also decreased by 14.71% following the intervention in patients with chronic stroke” [7]. This is related to the findings of source [3]’s study of 20 elderly individuals, who collectively saw a decrease of 26.6±16.1% in metabolic cost of transport, and a decrease of 4.24±2.57% in required oxygen consumption [3]. Reductions in these categories are integral in allowing individuals with impaired gaits to walk for longer periods without facing extreme fatigue.

By applying the proper torques at the correct times, HEs are also capable of encouraging muscle activation while simultaneously making up for the possible deficiencies that the wearer may have. Because of this ability, HEs can prevent muscle atrophy and promote continuous growth if needed to be worn for extended periods [2]. This is incredibly helpful for both groups of individuals who possibly benefit from using an HE: neurological disorder patients and the elderly. If HEs are eventually developed enough to become widespread household medical devices, the mobility of these impaired individuals will receive incredible benefits. For individuals with preexisting injuries, HEs also show extreme promise. It is not only the elderly and stroke survivors who could benefit from modern gait rehabilitation but younger people as well. Using a wearable robotic device, individuals with long-term disabilities/injuries could relearn their old walking patterns. HEs can redistribute loads across the lower body limbs, relieving pressure without compounding existing injuries.

Safety

Considering that stroke and other neurologically damaged individuals are the primary recipients of HEs, ensuring the wearer's safety is arguably the most important measure. When imagining an HE in use, the potential consequences of a control system malfunction during active use represent a serious safety concern. These risks have been carefully considered by the research community, and, as documented in sources [4], [6], [7], the implementation of adaptive control systems enabling real-time monitoring and adjustment renders such failures highly unlikely. By processing the real-time data constantly getting fed through the HE's sensors, the HE can adjust the level of resistance being provided [3].

Structured testing protocols are the primary mechanism through which safety concerns are identified and addressed. Unfortunately, as previously mentioned, HEs are still early in their development process, so limited testing has been completed. This observation motivates a central concern of this review – the potential adverse effects of continued use of an HE. As minimal testing has been completed in field studies and clinical trials, there exists no conclusive evidence to confirm whether HEs are 100% safe for daily integration into the wearer's lives [1]. Thus, we must examine the potential side effects of prolonged use of HEs to determine the safety concerns that arise. Sources [1] and [8] agree that more measures must be taken to ensure the efficacy of these HEs that are being designed.

Adverse Effects

As with any emerging technology, several engineering and clinical challenges remain to be addressed in mobility-assistive robotics. While numerous studies have evaluated the efficacy of hip exoskeletons for gait rehabilitation, comparatively little work has examined the potential drawbacks that users may face with continued HE use. As a literature review, source [1] takes a deeper look into the rate of adverse events (AEs) that occur within similar trials to those previously mentioned. While their review focuses on examining the AEs of stationary gait trainers, the results are similar enough to those of HEs to be significant. Of the 50 studies they examined, 18 of those had reports of AEs [1]. Of those AEs, the most common were soft-tissue issues, followed by musculoskeletal problems. Of the stated AEs, the most reported were “skin reddening, skin lesions, skin abrasions, a blood blister, chafing, skin irritation due to electromyography (EMG) electrodes, and bruises” [1].

Source [2] also reported adverse events observed during their study. Of the 738 training sessions their subjects completed, there were 34 reported AEs. Of those 34, “only 6 were determined to be device related” [2]. For most of source [2]’s AEs, the only reports were falls while the subject was mid-trial. Consistent with most other studies examined (excluding source [1]) in this review, the rate of adverse events and potential side effects was either unreported or only superficially addressed. Source [8] had similar findings to source [1] in that they saw a decrease in step length, an increase in step width, and an increase in gait variability. However, no attention was paid to actual side effects – outcomes such as bruising, muscular fatigue, or psychological effects [8]. While sources [2], [3], [4], [5], [7], [8] all considered the cardiovascular impacts of wearing an exoskeleton, none examined the potential drawbacks on strain rate that could come with constant use of an HE.

As mentioned above, there has been minimal work to acknowledge the psychological strain that HEs could place on users. As individuals with impaired gaits continue to use HEs in their gait therapy, they could potentially develop a reliance on the device. This could create some mental challenges, as the individual may not have the same confidence in their walking abilities if it was restored in a manner that wasn’t entirely their doing. This was not explored in any reviewed sources, but it poses a threat that should be taken seriously. Source [4] also tracked AEs as they completed their study, but similar to source [2], they only accounted for “…aggravation of knee pain or skin problems (redness or abrasion due to contact with the equipment)…,” of which they found none [4]. Future studies must take a more comprehensive approach to examining the potential negatives of wearing an HE. Given limited data, it is not possible to conclude whether any adverse effects occur at a statistically significant rate.

Conclusion

While hip exoskeletons present groundbreaking advancements in gait rehabilitation, they remain in early stages of development. Existing research consistently demonstrates substantial improvements across multiple gait parameters in individuals with mobility impairments. "[Walking] speed, step length, stride length, cadence, and oxygen consumption" are a few of the parameters that can directly benefit from using HEs in tandem with traditional rehabilitation therapies [7]. These robotic devices are designed to assist those with preexisting conditions that affect their daily mobility. However, a notable gap exists in the literature regarding the potential adverse effects of prolonged HE use. Soft tissue complications, musculoskeletal strain, and psychological dependency represent a broader class of potential concerns arising from HE use, yet each has received only minimal investigation. Addressing these gaps through rigorous longitudinal study would position hip exoskeletons as clinically validated tools capable of meaningfully improving daily mobility for a wide population.

References

Sources

[1] J. Bessler, G. B. Prange-Lasonder, R. V. Schulte, L. Schaake, E. C. Prinsen, and J. H. Buurke, “Occurrence and Type of Adverse Events During the Use of Stationary Gait Robots—A Systematic Literature Review,” *Front. Robot. AI*, vol. 7, Nov. 2020, doi: 10.3389/frobt.2020.557606.

[2] R. Macaluso *et al.*, “Safety & efficacy of a robotic hip exoskeleton on outpatient stroke rehabilitation,” *Journal of NeuroEngineering and Rehabilitation*, vol. 21, no. 1, p. 127, Jul. 2024, doi: 10.1186/s12984-024-01421-x.

[3] E. Martini *et al.*, “Gait training using a robotic hip exoskeleton improves metabolic gait efficiency in the elderly,” *Sci Rep*, vol. 9, no. 1, p. 7157, May 2019, doi: 10.1038/s41598-019-43628-2.

[4] K. Koseki *et al.*, “Gait Training Using a Hip-Wearable Robotic Exoskeleton After Total Knee Arthroplasty: A Case Report,” *Geriatric Orthopaedic Surgery & Rehabilitation*, vol. 11, p. 2151459320966483, Oct. 2020, doi: 10.1177/2151459320966483.

[5] B. Abdikadirova, M. Price, J. M. Jaramillo, W. Hoogkamer, and M. E. Huber, “Gait Adaptation to Asymmetric Hip Stiffness Applied by a Robotic Exoskeleton,” *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, vol. 32, pp. 791–799, 2024, doi: 10.1109/TNSRE.2024.3354517.

[6] S.-H. Hsu, C. Changcheng, H.-J. Lee, and C.-T. Chen, “Design and Implementation of a Robotic Hip Exoskeleton for Gait Rehabilitation,” *Actuators*, vol. 10, no. 9, Art. no. 9, Sep. 2021, doi: 10.3390/act10090212.

[7] M. Daliri *et al.*, “Powered single hip joint exoskeletons for gait rehabilitation: a systematic review and Meta-analysis,” *BMC Musculoskeletal Disorders*, vol. 25, no. 1, p. 80, Jan. 2024, doi: 10.1186/s12891-024-07189-4.

[8] J.-H. Park, S. Kim, M. A. Nussbaum, and D. Srinivasan, “Effects of back-support exoskeleton use on gait performance and stability during level walking,” *Gait & Posture*, vol. 92, pp. 181–190, Feb. 2022, doi: 10.1016/j.gaitpost.2021.11.028.

Annotated Bibliography

[1] J. Bessler, G. B. Prange-Lasonder, R. V. Schulte, L. Schaake, E. C. Prinsen, and J. H. Buurke, "Occurrence and Type of Adverse Events During the Use of Stationary Gait Robots—A Systematic Literature Review," *Front. Robot. AI*, vol. 7, Nov. 2020, doi: 10.3389/frobt.2020.557606.

This literature review reviews the possible adverse effects of robot-assisted gait training (RAGT) but focuses mainly on stationary trainers. The authors searched various scientific databases for studies on RAGT, but there is a lack of information. The results reviewed 50 studies involving 985 subjects. 18 studies reported side effects (36%). Skin irritation was the most frequent effect, followed by joint pain and blood pressure changes.

[2] R. Macaluso *et al.*, "Safety & efficacy of a robotic hip exoskeleton on outpatient stroke rehabilitation," *Journal of NeuroEngineering and Rehabilitation*, vol. 21, no. 1, p. 127, Jul. 2024, doi: 10.1186/s12984-024-01421-x.

This article reviewed safety concerns regarding the impacts of an exoskeleton used by stroke patients for gait rehabilitation. It reviewed the use of the Samsung Gait Enhancing and Motivating System-Hip (GEMS-H) and concluded that it was mostly safe with a low rate of adverse effects.

[3] E. Martini *et al.*, "Gait training using a robotic hip exoskeleton improves metabolic gait efficiency in the elderly," *Sci Rep*, vol. 9, no. 1, p. 7157, May 2019, doi: 10.1038/s41598-019-43628-2.

This article discusses the Active Pelvis Orthosis (APO) as a tool for improving gait efficiency in elderly subjects. The study showed a large decrease in the Metabolic Cost of Transport (MCoT) for subjects who wore the APO. The APO's torque profile promoted eccentric activation of the hip extensors, which benefits elderly folks.

[4] K. Koseki *et al.*, "Gait Training Using a Hip-Wearable Robotic Exoskeleton After Total Knee Arthroplasty: A Case Report," *Geriatric Orthopaedic Surgery & Rehabilitation*, vol. 11, p. 2151459320966483, Oct. 2020, doi: 10.1177/2151459320966483.

This scientific article reviews a case study on using a hip exoskeleton to improve gait efficiency after total knee arthroplasty (TKA). A 76-year-old woman who recently had TKA in both knees was subjected to 18 sessions of 20 minutes of use of the Honda Walking Assist (HWA). Results found no adverse effects and knee function and walking ability greatly improved. Compared to a control group undergoing conventional therapies, the patient showed faster improvements in mobility.

[5] B. Abdikadirova, M. Price, J. M. Jaramillo, W. Hoogkamer, and M. E. Huber, “Gait Adaptation to Asymmetric Hip Stiffness Applied by a Robotic Exoskeleton,” *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, vol. 32, pp. 791–799, 2024, doi: 10.1109/TNSRE.2024.3354517.

This article was written and worked on by Professor Huber in the HRSL at UMass. It primarily focuses on how subjects react to asymmetric stiffness applied by the exoskeleton while in use. 13 subjects with unimpaired gaits walked on a treadmill and received different stiffness inputs from the exoskeleton. Conclusions found that forced asymmetry resulted in similar results to “split-belt” treadmill training - a different gait rehabilitation method.

[6] S.-H. Hsu, C. Changcheng, H.-J. Lee, and C.-T. Chen, “Design and Implementation of a Robotic Hip Exoskeleton for Gait Rehabilitation,” *Actuators*, vol. 10, no. 9, Art. no. 9, Sep. 2021, doi: 10.3390/act10090212.

This article presents a design for a hip exoskeleton. The design features 4 degrees of freedom movement, focusing on flexion/extension and abduction/adduction. The exoskeleton utilizes a Linear Extended State Observer (LESO)-based Fast Terminal Sliding Mode Controller (FTSMC) for control which allows for easy control without extreme models. Various tests were performed to examine the user walking, with the primary result being the reduction of energy expenditure.

[7] M. Daliri *et al.*, “Powered single hip joint exoskeletons for gait rehabilitation: a systematic review and Meta-analysis,” *BMC Musculoskeletal Disorders*, vol. 25, no. 1, p. 80, Jan. 2024, doi: 10.1186/s12891-024-07189-4.

This article is a systematic review and meta-analysis that compares the use of a hip exoskeleton to traditional methods for gait rehabilitation. The authors reviewed previous studies to analyze the results on parameters like speed, step length, stride length, cadence, and oxygen consumption. They found that speed, step length, and oxygen consumption all improved.

[8] J.-H. Park, S. Kim, M. A. Nussbaum, and D. Srinivasan, “Effects of back-support exoskeleton use on gait performance and stability during level walking,” *Gait & Posture*, vol. 92, pp. 181–190, Feb. 2022, doi: 10.1016/j.gaitpost.2021.11.028.

This study reviews the adverse effects of a back-support exoskeleton (BSE) on gait. The BSE supplies different levels of external torque to assist with walking. The study reviewed effects such as step length, step width, cycle time, swing time, double support time, and gait variability. It concluded that while a BSE can offer some benefits in terms of stability, it can also lead to less efficient walking patterns and gait variability.